

Biomarker Interaction Implementation: Architecture Comparison

Contents

Empirical Comparison	3
Architecture B: Differential Correlation (MVN)	3
Architecture A: Direct Mean Moderation	3
Why the Architectures Diverge	3
Architecture A: Biomarker scaling is proportional	3
Architecture B: Correlation erosion compounds mean blurring . .	4
Biological Assumptions	4
When Architecture A is appropriate	5
When Architecture B is appropriate	5
Selecting the appropriate architecture	5

List of Tables

1	Architecture B power by design and carryover half-life	3
2	Architecture A power by design and carryover half-life	3

Empirical Comparison

Identical simulation configuration across both architectures: - Sample size: $N = 70$ - Biomarker correlation / moderation: $c.bm = 0.45$ - Analysis model: nlme::lme with corCAR1 and bm:Dbc interaction - Replicates: 50 per cell

Architecture B: Differential Correlation (MVN)

Power under carryover half-life variation:

Table 1: Architecture B power by design and carryover half-life

Design	t_half = 0	t_half = 0.5	t_half = 1.0
N-of-1	0.82	0.64	0.50
CO	0.40	0.40	0.32
OL+BDC	0.62	0.38	0.24

Power declines substantially with increasing carryover: - N-of-1: 68% relative loss - CO: 20% relative loss - OL+BDC: 61% relative loss

Architecture A: Direct Mean Moderation

Power under carryover half-life variation:

Table 2: Architecture A power by design and carryover half-life

Design	t_half = 0	t_half = 0.5	t_half = 1.0
N-of-1	0.74	0.72	0.68
CO	0.38	0.36	0.34
OL+BDC	0.62	0.58	0.54

Power is largely preserved under carryover: - N-of-1: 8% relative loss - CO: 11% relative loss - OL+BDC: 13% relative loss

Why the Architectures Diverge

Architecture A: Biomarker scaling is proportional

Under direct mean moderation, carryover reduces the magnitude of the drug exposure variable D during off-drug periods, but the biomarker moderation coefficient (β_{bm}) remains a fixed population parameter. The interaction

(bm * D) has reduced variance because D has less contrast, but the signal-to-noise ratio degrades only modestly since the noise structure is independent of carryover.

Mathematically, the expected outcome difference between two participants with biomarker values b1 and b2, both on drug, is:

$$E[Y|b_1, D = 1] - E[Y|b_2, D = 1] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D$$

This difference is constant regardless of carryover: if D is replaced by D * carryover_factor, the difference scales by the same factor, but the ratio is preserved.

Architecture B: Correlation erosion compounds mean blurring

Under the MVN approach, carryover operates on two levels simultaneously:

1. **Mean blurring:** The BR means during off-drug periods are inflated by residual carryover, making them resemble on-drug BR means. This reduces the treatment contrast that the analysis model uses to distinguish drug effect from no-drug.
2. **Correlation erosion:** The BM-BR correlation during off-drug periods decays exponentially with time since discontinuation:

$$\text{Cor}(BM, BR|t_{sd}) = c_{bm} \cdot e^{-\lambda t_{sd}}$$

This is not merely a statistical modeling choice – it is a structural consequence of the DGP. The correlation between biomarker and BR reflects the degree to which the drug effect is active; as the drug washes out, the correlation weakens because the drug-mediated component of BR variance shrinks relative to the drug-independent component.

The analysis model's `bm:Dbc` interaction coefficient depends on the covariance between biomarker and Dbc-weighted outcomes. Under Architecture B, this covariance is being attacked from both sides: the treatment contrast in Dbc is compressed, and the biomarker-BR correlation that generates the covariance signal is simultaneously decaying. This dual effect produces the 40-60% power losses observed in the simulations.

Biological Assumptions

The choice between architectures reflects fundamentally different assumptions about the mechanism by which the biomarker moderates treatment response.

When Architecture A is appropriate

Architecture A (direct mean moderation) is appropriate when the biomarker governs the **magnitude of the drug’s biological effect** for each individual participant. Under this mechanism, the drug effect for a participant is proportional to their biomarker value: higher biomarker predicts larger treatment response.

Example: A pharmacogenetic biomarker that determines hepatic metabolism capacity. Participants with high enzyme expression metabolize the drug more rapidly, but when the drug is present (regardless of whether it has completely washed out between doses), they experience proportionally larger biological response. The interaction is a property of each individual’s biology.

When Architecture B is appropriate

Architecture B (differential correlation, MVN) is appropriate when the biomarker’s predictive value emerges from **shared variance** with a drug-responsive physiological component, and this shared variance decays as the drug is eliminated.

Example: A dynamic biomarker that reflects current drug pathway engagement (e.g., active drug metabolite concentration). The biomarker and drug response are jointly determined by the same underlying pathway state. During off-drug periods, both the biomarker and the drug-responsive component of BR decay in parallel, so their correlation weakens. The interaction is a property of the joint response to the drug, not an inherent property of the individual.

Selecting the appropriate architecture

For simulation studies of predictive biomarker-moderated treatment effects, the choice should be driven by the clinical hypothesis:

- **Baseline biomarker predicts intrinsic drug responsiveness:** Use Architecture A (direct mean moderation)
- **Biomarker reflects current drug pathway engagement:** Use Architecture B (differential correlation)

The power implications are substantial: the choice between architectures determines whether carryover substantially erodes power (Architecture B) or whether power is largely preserved (Architecture A). This choice must be made based on clinical reasoning, not statistical convenience.

Rendered on 2026-04-08 at 09:47 PDT. Source: ~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/docs/21-architecture-comparison-summary.md